PROJECT BRIEF · BUSINESS PROBLEM STATEMENT

Superstore Sales Analysis

The business questions behind a four-year look at revenue, profit, customers, discounts, shipping and seasonality, plus a set of advanced SQL problems built to test database skill.

39	10	7	4
business questions	advanced SQL problems	areas of the business	years of sales data

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Where does Superstore make money, and where does it lose it?

WHAT THIS PROJECT IS ABOUT

The Business Problem

A retailer with four years of orders and one question it cannot yet answer.

This project is built around **39 business questions**. Everything else, from preparing the data to the analysis, the SQL work and the charts, exists to answer them. Alongside those questions sits a second set of **10 advanced SQL problems** that put the database side of the project under real pressure.

BACKGROUND

Superstore is a U.S. retailer that sells **Furniture, Office Supplies and Technology** to **Consumer, Corporate and Home Office** customers across four regions. It holds four years of order data, from 2009 to 2012, yet there is no clear picture of one simple thing: **where does the business make money, and where does it lose it?**

OBJECTIVE

To find out which regions, products, customers and discount levels make money and which lose it, and to say plainly what the business should sell more of, stop pushing, or change.

SCOPE

The analysis works only from the historical order data. It describes what happened, explains why where the data allows it, and turns the results into recommendations the business can act on. It does not try to forecast beyond what the questions ask for.

AT A GLANCE

39	7	10	4
business questions	business categories	advanced SQL problems	years of sales data

QUESTIONS BY CATEGORY

Category	Name	Questions	Range
Category 1	**Overall Business Performance**	5	Q01 – Q05
Category 2	**Regional Performance**	6	Q06 – Q11
Category 3	**Product & Category Analysis**	7	Q12 – Q18
Category 4	**Customer Analysis**	6	Q19 – Q24
Category 5	**Discount & Profitability Analysis**	5	Q25 – Q29
Category 6	**Shipping & Operations Analysis**	5	Q30 – Q34
Category 7	**Time Series & Trend Analysis**	5	Q35 – Q39
Total		**39**	**Q01 – Q39**

PART 1

The 39 Business Questions

Each category opens with the concern behind it, followed by the questions to answer.

CATEGORY 1 — 5 questions

Overall Business Performance

> *"Give me a snapshot of how the business is doing overall"*

Q01	What is the total revenue, total profit, total number of orders, and total quantity sold across all years?
Q02	What is the overall profit margin of the business? (Profit / Sales × 100)
Q03	How has total revenue trended month by month across all years? Is the business growing?
Q04	Which year was the most profitable and which was the worst?
Q05	What is the average order value?

CATEGORY 2 — 6 questions

Regional Performance

> *"We want to know which parts of the country are making us money and which are costing us"*

Q06	Which region generates the highest total revenue?
Q07	Which region generates the highest total profit?
Q08	Which region has the best profit margin percentage?
Q09	Which state is the most profitable and which state is losing us money?
Q10	Which city generates the most revenue?
Q11	Are there any states where we are consistently losing money despite decent sales?

CATEGORY 3 7 questions

Product & Category Analysis

> *"We need to know what to sell more of and what to stop pushing"*

Q12	Which category (Furniture, Office Supplies, Technology) generates the most revenue?
Q13	Which category is the most profitable and which has the worst profit margin?
Q14	Which sub-category has the highest profit and which has the lowest (or negative) profit?
Q15	What are the top 10 best-selling products by revenue?
Q16	What are the top 10 most profitable products?
Q17	Are there products with high sales but negative profit, meaning we are selling at a loss?
Q18	Which sub-categories are dragging down overall profitability?

CATEGORY 4 6 questions

Customer Analysis

> *"Who are our best customers and which customer type is most valuable?"*

Q19	Which customer segment (Consumer, Corporate, Home Office) generates the most revenue?
Q20	Which customer segment is the most profitable?
Q21	Who are the top 10 customers by total revenue?
Q22	Who are the top 10 customers by total profit?
Q23	How many unique customers does the business have?
Q24	Are there customers who generate high revenue but negative profit?

CATEGORY 5 5 questions

Discount & Profitability Analysis

> *"We suspect our discount strategy is hurting profits. Prove or disprove this"*

Q25	What is the relationship between discount percentage and profit? Does higher discount mean lower profit?
Q26	Which sub-categories receive the highest average discounts?
Q27	Are we giving discounts on products that are already unprofitable?
Q28	What is the profit margin at different discount levels (0%, 1-20%, 21-40%, 40%+)?
Q29	Which orders had the deepest discounts and what was the profit on those orders?

CATEGORY 6 5 questions

Shipping & Operations Analysis

> *"We want to understand our shipping patterns and whether they affect profitability"*

Q30	Which shipping mode is used most frequently?
Q31	What is the average shipping time (days between order date and ship date) per shipping mode?
Q32	Which shipping mode is associated with the highest profit?
Q33	Are there orders where shipping took unusually long? How many?
Q34	Which region has the fastest average shipping time?

CATEGORY 7 5 questions

Time Series & Trend Analysis

> *"We want to understand seasonality and plan for next year"*

Q35	Which month of the year consistently has the highest sales across all years?
Q36	Which quarter performs best in terms of revenue and profit?
Q37	Is there a seasonal pattern? Do sales peak at certain times every year?
Q38	Which month has the lowest sales and should we run promotions during that period?
Q39	What is the year-over-year growth rate in revenue and profit?

PART 2

The 10 Advanced SQL Problems

A second set of questions, written to be solved in a real database.

The 39 business questions can be answered in Python. These ten are different. They are written as problems that an analyst meets on the job and that a hiring manager uses to separate someone who knows SQL syntax from someone who thinks in SQL. They will be solved in **PostgreSQL**, managed through **DBeaver**, against a properly structured relational database rather than a flat file.

Each problem is built around a technique that basic queries cannot handle: ranking with ties, running and moving calculations, filtering on ranked results, period-over-period growth, percentile-based customer flags, and segmentation by behaviour. Together they show an understanding of how data is modelled, how window functions and CTEs behave, and how to write queries that stay correct when the data gets awkward.

S01 Customer Lifetime Value Ranking

Rank all customers by the total profit they have generated, using RANK() and DENSE_RANK() side by side to show how each one treats ties.

SKILL TESTED: RANKING FUNCTIONS

S02 Running Total and Moving Average of Monthly Revenue

Calculate cumulative revenue over time next to a 3-month moving average, to smooth out volatility and expose the underlying trend.

SKILL TESTED: WINDOW FRAMES

S03 Top 3 Products by Profit Within Each Category

Find the three most profitable products in every category with a ranked CTE, working around the fact that window functions cannot be filtered directly in a WHERE clause.

SKILL TESTED: CTES, PARTITIONED RANKING

S04 Month-over-Month and Year-over-Year Revenue Growth

Calculate both MoM and YoY percentage revenue change in a single query, with protection against division by zero.

SKILL TESTED: PERIOD COMPARISON, SAFE DIVISION

S05 High-Value, High-Risk Customers

Identify customers whose total sales fall in the top 20% but whose average profit margin sits below the company-wide average, flagging big accounts that are quietly unprofitable.

SKILL TESTED: PERCENTILES, BENCHMARK COMPARISON

S06 Sub-Category Profitability Rank Within Each Region

Rank every sub-category's profit margin inside its own region, to surface the best and worst performer in each one.

SKILL TESTED: PARTITIONED RANKING

S07 **Discount Threshold Where Profitability Breaks Even**

Group orders into discount bands and find the exact band where average profit margin flips from positive to negative.

SKILL TESTED: BANDING, CONDITIONAL LOGIC

S08 **Customer Order Frequency and Retention Segments**

Classify customers as One-Time, Occasional or Frequent based on order count, then measure how much profit each segment contributes.

SKILL TESTED: SEGMENTATION, AGGREGATION

S09 **Shipping Performance vs. Category Benchmark**

Identify every order that shipped slower than the average for its product category, and rank how far over the benchmark each one ran.

SKILL TESTED: BENCHMARKING, RANKING

S10 **Executive KPI Snapshot with Year-over-Year Comparison**

Produce revenue, profit, margin and order count for the current year against the previous year, side by side, in a single query.

SKILL TESTED: CONDITIONAL AGGREGATION

WHY THE OVERLAP

A few of these problems touch the same ground as the business questions on purpose: Q03 and S02 (revenue trend), Q22 and S01 (customer profit), Q28 and S07 (discount levels), Q39 and S04 (growth). Working the same idea in two tools makes it possible to check one answer against the other, and shows the logic holds up whichever tool is used.

PART 3

Approach and Technology

How the work is organised, and the tools used at each stage.

TECHNOLOGY STACK

Tool	Role in the project
Python	Core language for data preparation, analysis and workflow
Pandas	Loading, cleaning, reshaping and aggregating the order data
NumPy	Numerical work and efficient calculations
Statistics	Descriptive statistics, correlation, outlier detection and distribution analysis behind the findings
Matplotlib	Charts built to a consistent, controlled standard
Seaborn	Statistical visuals for distributions and relationships
SQL	Querying, ranking, windowing and aggregating inside a database
PostgreSQL	The relational database that holds the structured data
DBeaver	Database client for designing tables and writing queries

PROJECT PHASES

#	Stage	What is done	Tools	Purpose
1	**Project setup**	Folders, environment and imports	Python	A clean starting point
2	**Data understanding**	Load the data, look through it, understand how it is structured	Pandas	Know the data before using it
3	**Data cleaning**	Fix data types, missing values and columns	Pandas	Make the data trustworthy
4	**Exploratory analysis**	Distributions, outliers and relationships	Pandas + NumPy	Spot patterns early
5	**Business analysis**	Answer the 39 business questions	Pandas + NumPy	The main analysis
6	**Visualisation**	A chart for every finding	Matplotlib + Seaborn	Show the results clearly
7	**SQL analysis**	Build the database and solve the 10 advanced SQL problems	PostgreSQL + DBeaver	Show database and SQL understanding
8	**Dashboard**	An interactive view of the results	Power BI	Let others explore the data
9	**Insights and recommendations**	Write up the findings and what to do about them	Report	Explain it in business terms
10	**Documentation**	Publish the project with clear notes	Git + GitHub	Make the project easy to follow

HOW EACH QUESTION IS HANDLED

Charts are made as each question is answered, not left to the end. Every business question follows the same pattern, so the notebook reads cleanly from top to bottom:

1	2	3	4	5
Question	Code	Result	Chart	One-line insight

The SQL problems follow their own rhythm: state the problem, write the query, check the output against what the data should produce, and note anything that made the query harder than it first looked.

PLANNED DELIVERABLES

Cleaned dataset	Checked, documented and ready to analyse
Analysis with charts	Every business question answered, with a chart and a short insight
Database and SQL scripts	A PostgreSQL database with the ten advanced problems solved and commented
Dashboard	An interactive view that others can filter and explore
Written report	Findings and recommendations in plain language
GitHub repository	Everything documented in one place

> **WHY THE QUESTIONS COME FIRST**
>
> The purpose of this document is to fix the questions before any answers exist. The 39 business questions and the 10 SQL problems define what the project has to prove, and every later decision about method, charts and presentation follows from them.